{Nano-Innovator-Grant Proposal} Nanosecond, multi-addressable, molecular gates made from self-replenishing, photocleavable lipid-monolayers

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Abstract

A molecular gate that possesses a sub-microsecond switching time, near-perfect molecular gating, ease of fabrication, and a long functional lifespan has not yet been fabricated.

Many devices in the literature attempt to achieve fast switching times by reversibly electrowetting the gates[1], reversibly mineralizing the gates[2], or by using FET-like gating[3], but these fail to achieve sub-millisecond switching times.

Other methods try to exploit the slip conditions which occur at atomically smooth, hydrophobic channel interiors to create molecular, one-way flows through the gates, eliminating reverse molecular diffusion. Carbon nanotubes[4] and biological pores[5] are able to achieve this level of smoothness, but carbon nanotubes are inherently difficult to process and biological nanopores tend to degrade within a few hours at room temperature[6]. Furthermore, they act only as effective flow reversal devices, not strictly as gates.

Piezoelectric pumps and piezoelectric latches switch at tens of kHz speeds and have long cycle lifetimes[7]. Nonetheless, they typically possess large footprints and so are not suitable for single molecule gating.

It may seem that no current gating methods satisfy all four criteria. It is clear, however, that a gating mechanism which possesses all of these attributes could lead to a new generation of chemical microreactors. Chemical solutions could be probed for the formation of products or the activity of a protein and immediately disturbed by a pulse of chemicals into the solution, thereby altering its reaction path live. Drug discovery, DNA synthesis, and synthetic chemistry would all benefit from this capability.

In this proposal, I present a molecular gate which excels in all four criteria - nanosecond switching, true molecular gating efficacy, fabrication using standard cleanroom protocols, and self-replenishing in nature.

Introduction

In a situation where molecular contamination is not tolerable, only atomically-sealed gates are a valid solution. One such atomically-sealed structure that stands out for its ease of fabrication and replenishability is the lipid-monolayer. Lipid-monolayers, also called Langmuir monolayers, appear at an oil-water interface (referred to as a meniscus), when lipids residing within the oil migrate to the meniscus. They protrude their hydrophilic heads into the aqueous solution and embed their hydrophobic tails into the non-polar solution. Each lipid molecule, by itself, does not have sufficient energy to dissolve into the aqueous solution[8], guaranteeing a molecular level of sterility for the aqueous phase. If the hydrophilic head is linked to its hydrophobic tail by a hydrolyzable, photocaged group, then upon illuminating it with ultraviolet light, the tail will cleave from the head, allowing the head, alone, to diffuse into the aqueous solvent[9][10][11]. Therefore, only head-groups active at the atomically thin oil-water interface while the UV pulse is present are allowed to diffuse into solution. UV pulses can be shortened to less than a nanosecond[12], allowing for very fine control over the chemical head-group injection. Following this logic, the total concentration of head-groups introduced into solution may be tuned by varying the pulse-width of the incoming light.

Methods

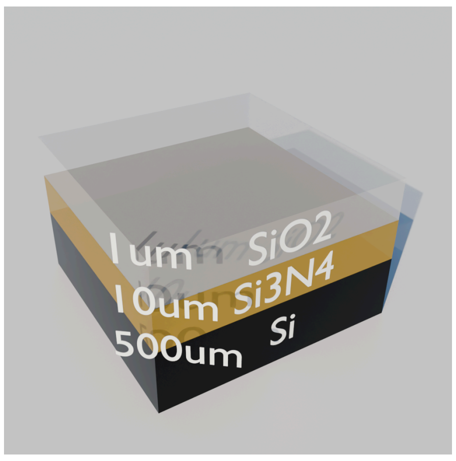
The main design of the microfluidic device which houses the mechanism is as follows. Two closed microfluidic channels reside on top of each other, as shown in Figure 1l. The roof of the lower channel and the floor of the upper channel are vertically separated by a 200 nm-thick thin film (Figure 1e.). Nanopores are etched through this thin film, creating a path for the liquids in the two channels to meet (Figure 1h.). When the two liquids meet, the interface they create is called a meniscus. Lipids, initially present in the hydrophobic solvent, form a lipid-monolayer at the meniscus. When light illuminates the specially engineered lipids, the head-groups cleave from the tail-groups and are released into the aqueous phase (Figure 3 and 5h).

Several design considerations must be taken into account when making this device. The fabrication of the fluidic channel is based on a sacrificial layer approach where SiO_2 is etched away via HF to develop the microfluidic channels[13]. The process flow was designed such that the only the SiO_2 in the microfluidic channels was left exposed to the HF after all the process steps were completed. The result of HF finding its way into the bulk of the SiO_2 would be the structural failure of the chip.

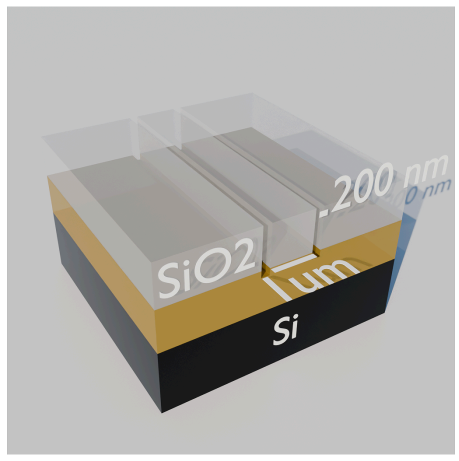
Emulsification and the positioning of the meniscus were also two engineering challenges that needed to be solved. The choice of using nanopores with both hydrophobic and hydrophilic domains solved both of these problems and will be discussed after the fabrication section.

Fabrication Process

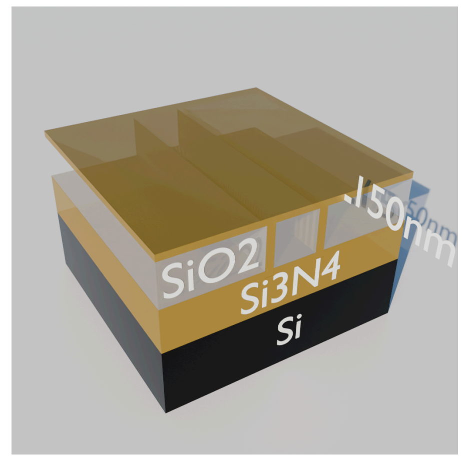
Figure 1



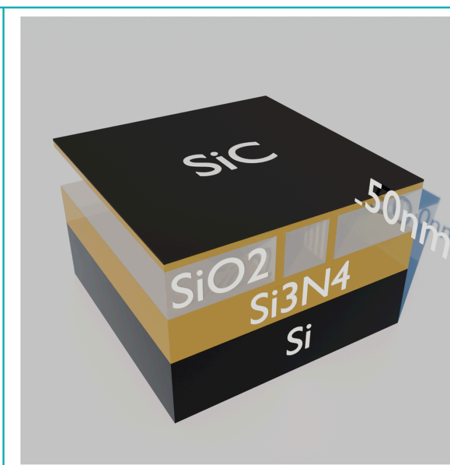
a. The initial stack is silicon-dioxide (SiO2/glass) on silicon-nitride (SiN) on silicon (Si).



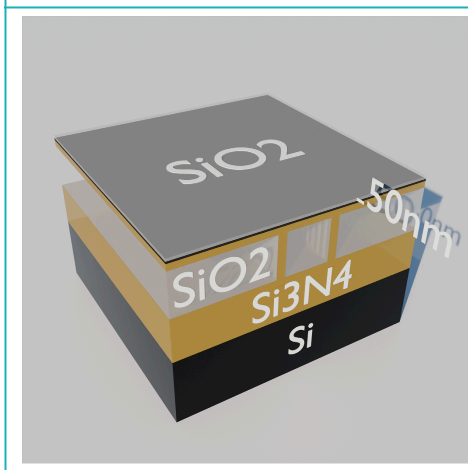
b. The lower channel is defined by RIE etching through the SiO2 on either side.



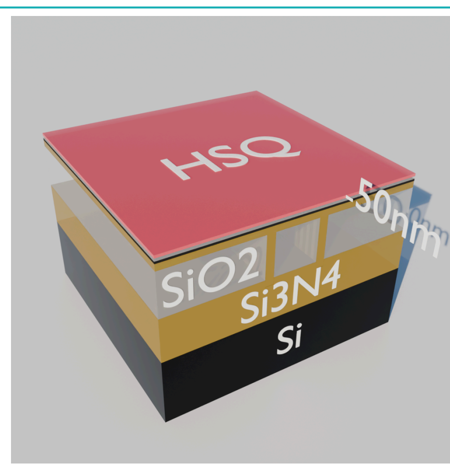
c. Silicon-nitride (hydrophilic) is deposited via PECVD, sealing off the lower channel.



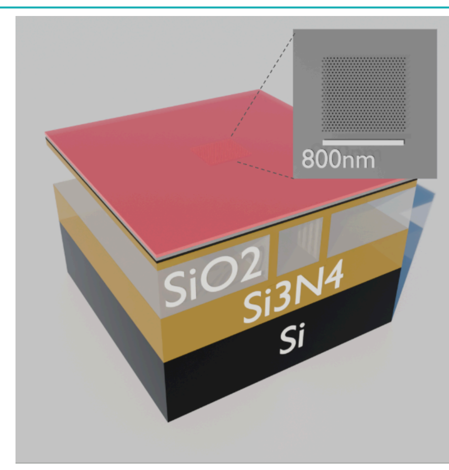
d. A thin layer of silicon-carbide (SiC, hydrophobic) is deposited via PECVD.



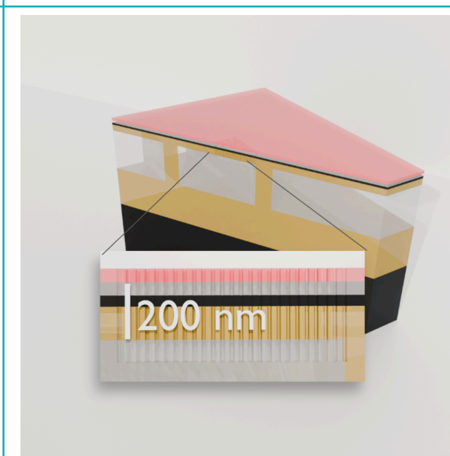
e. A thin layer of SiO2 is deposited to act as a ICP-RIE mask.



f. HSQ is deposited as a resist mask to pattern the glass mask.

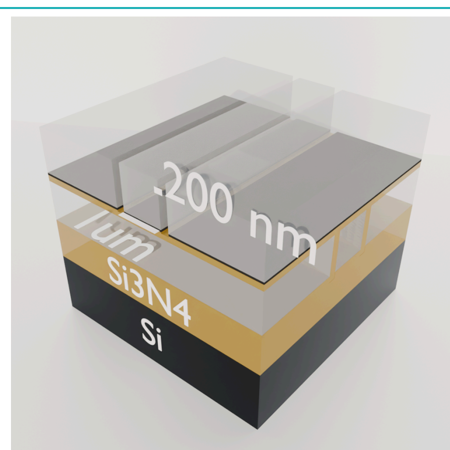


g. 20 nm diameter holes are patterned and etched through the glass mask[14], residing solely above the lower channel.

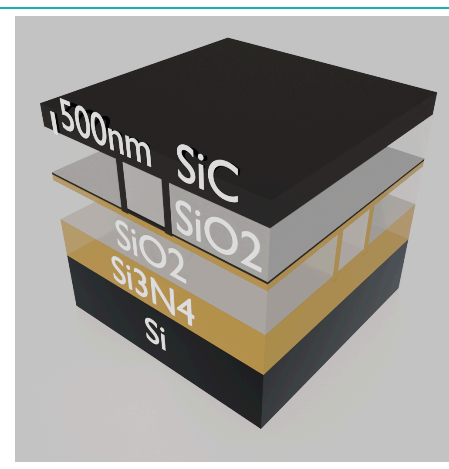


h. ICP-RIE is used to etch through the SiN layer (CHF_3/O_2)[15] and through the SiC layer (SF_6/O_2)[16], creating nanopores that pass through the SiN-SiC layers.

i. The HSQ is stripped and glass is deposited on top of the remaining glass mask.



j. The upper channel is defined by RIE etching through the SiO2 on either side. The upper channel directly crosses over the nanopore array.

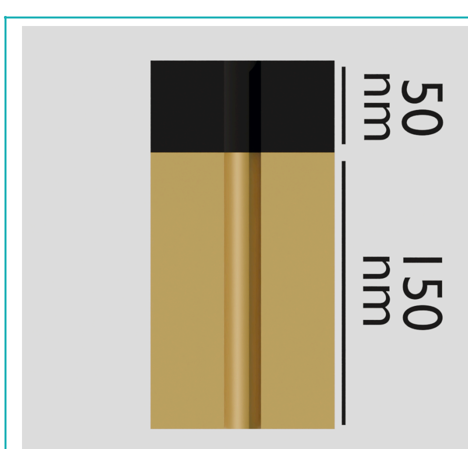


k. Silicon-carbide is deposited via PECVD, sealing off the upper channel.

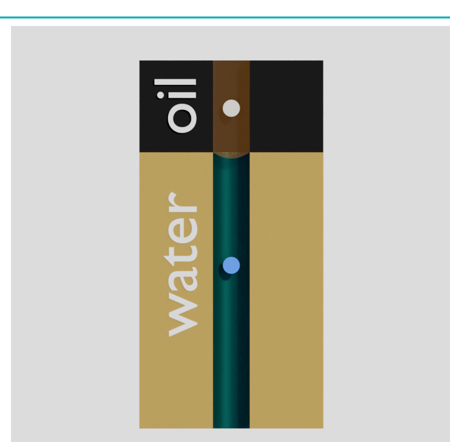
l. The device is soaked in HF for 24 hours, dissolving away the exposed SiO2 and forming the fluidic channels [13-1].

Design of the nanopore channels

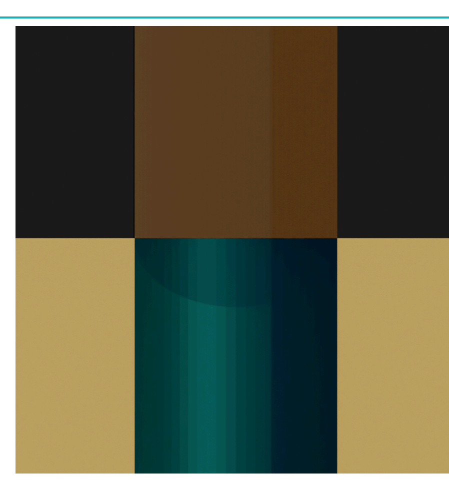
Figure 2



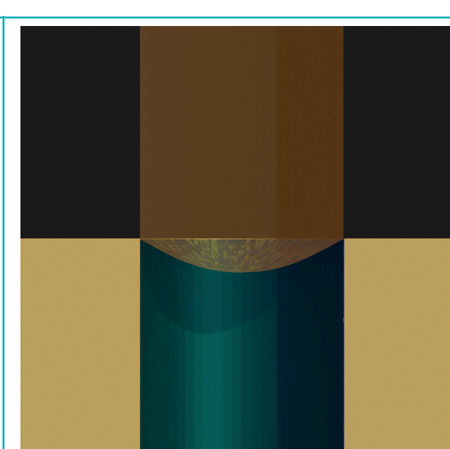
a. Zoomed in view of one particular nanochannel consisting of SiN (yellow) and SiC (black).



b. An aqueous and hydrophobic solvent meet at the sidewall transition between SiN and SiC [17].



c. Without a pressure gradient, the meniscus is nearly flat if the oil matches the SiC and the water matches the SiN well [17-1].



d. Higher pressure in the hydrophobic channel causes the meniscus to bulge downwards [18].

The nanopore channels (nanochannels) which connect the upper fluidic channel to the lower fluidic channel are designed such that the pressure fluctuations resulting from the flow in the upper channel are not felt at the level of the meniscus. This happens because the velocity profile of the flow in the main fluidic channel exponentially decreases deeper into the nanochannel [19]. This design choice will be useful later when emulsion prevention is discussed.

The position of the meniscus forms at the transition from SiN to SiC in the nanochannel wall due to capillary wetting forces[17-2] (see Figure 2b). Because properly post-processed, SiC is a hydrophobic material[20] while SiN is a hydrophilic material[21], the position of the meniscus will reach a thermodynamic equilibrium when the oil fully coats the SiC sidewalls and the water fully coats the SiN sidewalls. Although there may be a pressure difference between the two channels, the position of the meniscus will not be affected. This is due to the tiny radii of the channels requiring an unattainably large pressure difference to break the surface tension of the meniscus and force liquid from one side to the other[18-1]. The main effect of the pressure difference will be on the curvature of the meniscus which bulges away from the channel with the higher pressure[18-2].

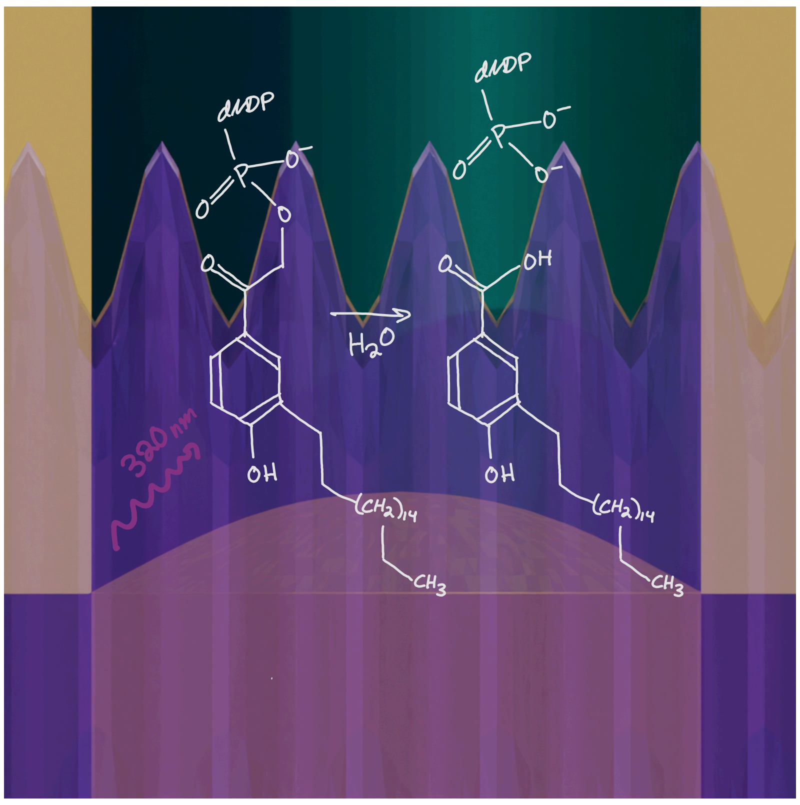
Design of the lipid

Emulsion Prevention

A lipid is made up of two parts: a hydrophobic tail and a hydrophilic head.
Wherever there is an oil-water interface and lipids are present in the oil, a monolayer of lipids will form at the interface. However, there is also a chance of an emulsion forming. An emulsion forms when enough energy - mechanical or thermal - is supplied to the monolayer to cause a droplet of oil, surrounded by lipids, to enter into the aqueous environment. This oil-monolayer complex is called an emulsion. It is thermodynamically stable because the hydrophobic tails stabilize the oil phase while the hydrophilic heads stabilize the liquid phase. However, with a reduction of flow fluctuations at the meniscus interface (as was already discussed) and a properly designed lipid, emulsions may be prevented.
The favorability of an emulsion forming is governed by a value called the packing parameter, P. P is defined as $\frac{V}{l*a_0}$, where V is the total volume of the lipid, l is the length of the tail, and a_0 is the cross-sectional area of the head-group. Typically, emulsions are favored to form when $P < \frac{1}{3}$, where lipids roughly represent a cone-like shape, but this equation breaks down when the cross-sectional area of the head-groups surpass the cross-sectional area of the lipid tails. In this regime, emulsions are unable to form because cones with too high an aspect ratio do not pack well enough to form a tight shell around an oil droplet[22].
Vesicles, emulsions, and micelles are prevented when charged head-groups form a large hydration shell that prevents lipids from packing closely without letting water through [23].
Because our our head-group in the synthesis section will be chosen to be dNTP, a massive and highly charged head with a large hydration shell, the chances of emulsions or micelles or vesicles forming are unlikely.

Photocleave chemistry

Figure 3



Under the illumination of 380 nm light, pHP undergoes a multistep reaction that releases the leaving group, in this case is a deoxyribonucleoside triphosphate, with water acting as a nucleophilep1.

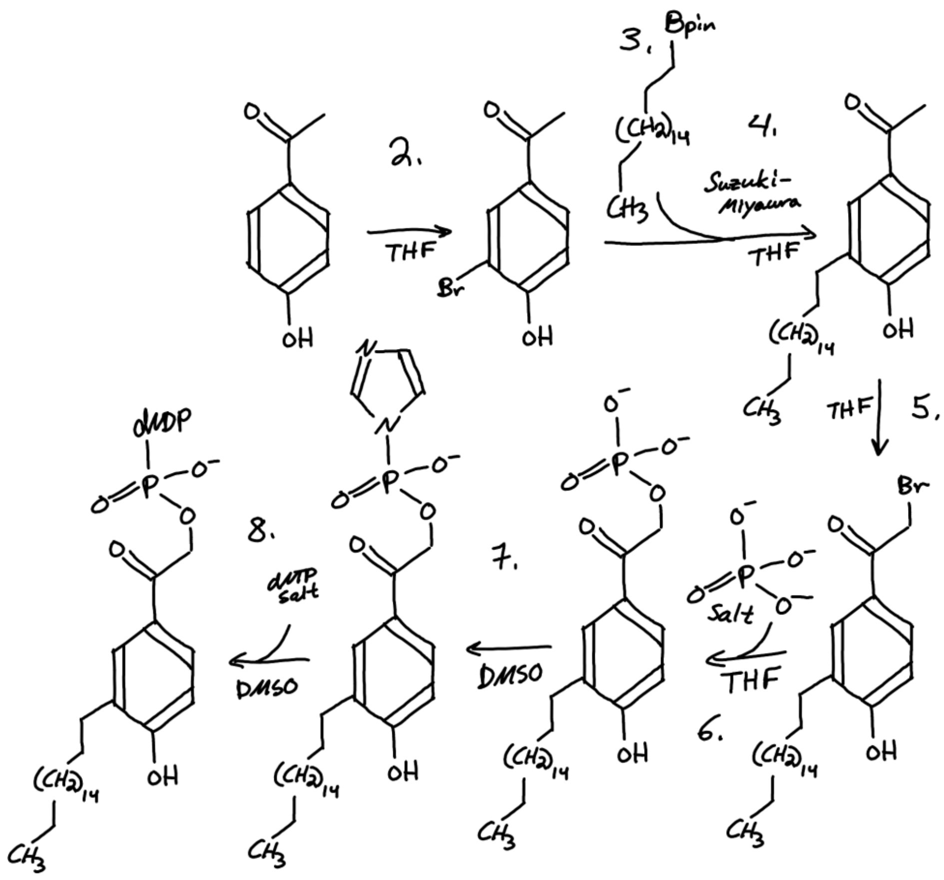
The main role of the lipid is to act as the gate through which chemicals are introduced to the aqueous phase. A lipid, by itself, has a very positive free energy of entering the aqueous phase, much larger than k_bT. This is due to the unfavorability of the water solvating the many hydrophobic tail groups. Therefore, the lipid head-group never diffuses into the aqueous phase by itself.
To control when the head-group dissociates, we may design the lipid so that the head is linked to the tail by a photocaging group, which, when activated, releases the head into solution. Once the head is released into solution, what is left of the lipid is no longer hydrophilic, so the remains sink back into the bulk of the oil solvent, allowing an intact lipid to take its place. The linker we will choose for this lipid is pHP, which requires the presence of water in its photocleaving mechanismp1. Extremely fast pulses of light will thereby release molecules only at the atomically thin oil-water interface, and there will be no concern of free head-groups within the bulk of the oil slowly

diffusing into the aqueous solution over time.
Therefore, we have successfully created a chemical gate which has nanosecond control over its dosage, is self-replenishing, and does not foul the fluidic channels because it is only a small molecule.

[p1]: Corrie, John E.T, et al. *Photoremovable Protecting Groups Used for the Caging of Biomolecules*. 17 Feb. 2005, pp. 1–94, https://doi.org/10.1002/3527605592.ch1. Accessed 27 Nov. 2025.

Synthesis of the photocleavable lipid

Figure 4



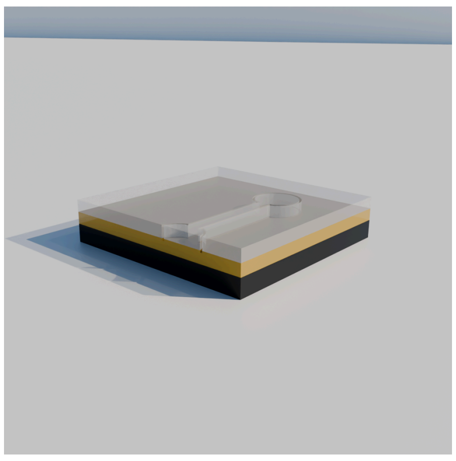
Synthetic pathway of 3'-octadecyl-(p-hydroxyacetophenyl) dNTP. 1. Core reagentss3[24][25] *2. Regioselective bromination*[26] *3. Miyaura borylation*[27] *4. Suzuki-Miyaura reaction*[28] *5. α-bromination*[29] *6. SN2 Substitutionp1 7. Imidazole pre-priming*[30] *8. dNTP salt coupling*[30-1]

The synthesis of the photocleavable lipid is done in nine relatively non-toxic steps, with the approximate NFPA health hazard rating per step being 2. The head-group chosen to be appended to pHP is any nucleoside triphosphate. The following details the reaction steps:

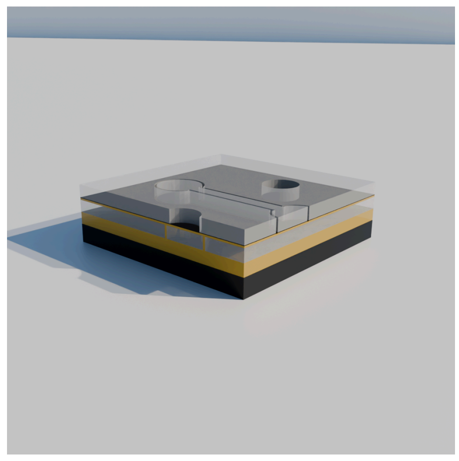
1. p-Hydroxyacetophenone (pHA) and 1-bromooctadecane (Br-C18) are purchased from a chemical supplier.
2. pHA is regioselectively 3'-brominated in THF forming pHA-Br and purified
3. Br-C18 is borylated in THF via a Miyaura borylation reaction forming Bpin-C18 and purified.
4. pHA-Br and Bpin-C18 are coupled together in THF to form C18-pHA via a Suzuki-Miyaura reaction, with C18 attached at the 3' position of pHA, and purified. It has been shown that ring substituents that keep the aromatic structure the same tend to have little effect on the photochemistry[31].
5. C18-pHA is converted into 3'-octadecyl-(p-hydroxyphenacyl) bromide (C18-pHP-Br) via an α-bromination of the pHA ketone with PHPB in THF and purified.
6. C18-pHP-Br is converted into C18-pHP-O-PO_3 via an SN2 nucleophilic substitution reaction with an ion-exchange resin salted phosphate in THF and purified.
7. C18-pHP-O-PO_3 is pre-primed to become C18-pHP-O-PO_2-imidazole by slowly dripping C18-pHP-O-PO_3 into a solution of DMSO containing a high concentration of imidazole and purified. Because the rate kinetics for this reaction is approximately first order in C18-pHP-O-PO_3 and second order in C18-pHP-O-PO_3 + C18-pHP-O-PO_2-imidazole, the likelihood of pHP-like dimers forming in solution is low, and their concentration may be neglected.
8. Ion-exchange-resin-salted dNTPs are added into C18-pHP-O-PO_2-imidazole in DMSO and is converted into C18-pHP-dNTP and purified.
9. C18-pHP-dNTP is diluted in n-octadecane.

Layout of the chip

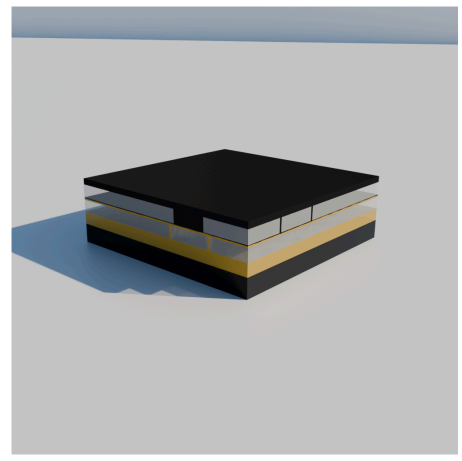
Figure 5



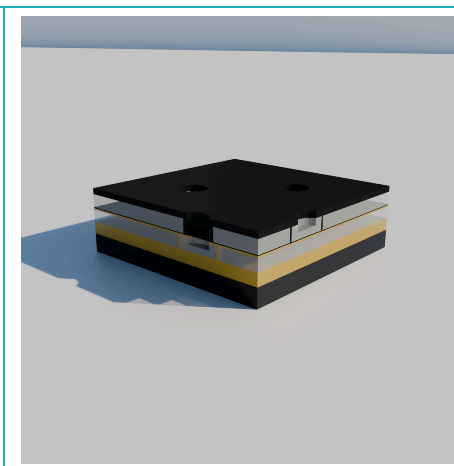
a. Crossectional view. Included in the first etch step are lower channel ports for the solvent.



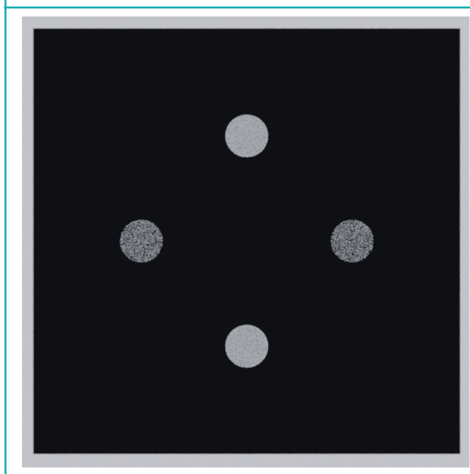
b. SiN is deposited followed by SiO2 deposition, followed by etching through the SiN to expose the lower channel port.



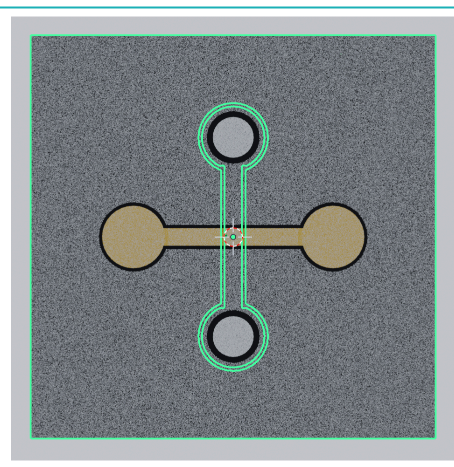
c. SiC is deposited via PECVD.



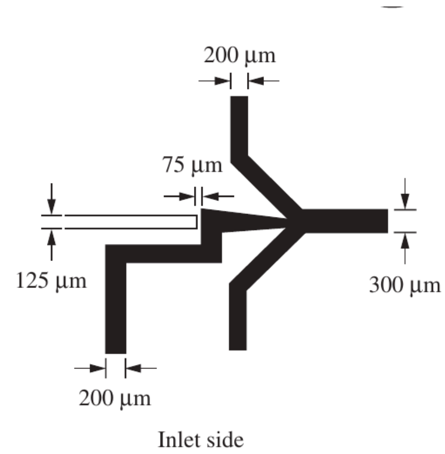
d. Etching is performed to expose both ports. HF will later dissolve the SiO2 in the exposed ports away.



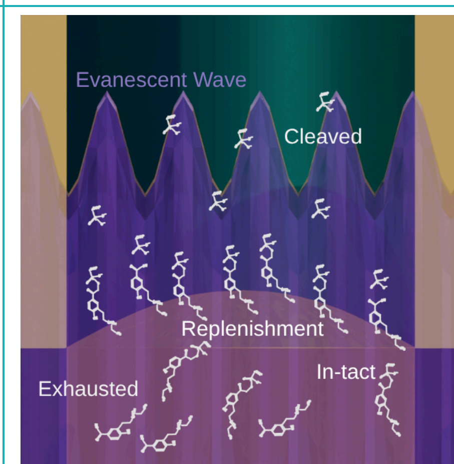
e. Top down view of chip architecture with upper channel ports (dark grey) and lower channel ports (light grey) shown.



f. X-ray view of the upper channel ports (yellow) and the lower channel ports (green).



g. The true version of this optofluidic device will contain a 90 degree turn for coupling light into the fluidic channel as is shown in this diagram from Chapter 3 of *Optofluidics*o1. The optical fiber is white, and the fluidic channels are black.



h. UV light from the fiber enters into the aqueous microfluidic channel and the evanescent tale overlaps with the monolayer, cleaving head groups off. New lipids come to replenish those that have lost their hydrophilicity.

Although the fabrication of the chip was generally detailed, the chip still needs to fulfill two more requirements. First, it needs in-ports and out-ports for both the upper and lower channel. Second, there needs to be a way to couple light into and out of the chip. The process of creating in and out ports is detailed in Figure 5a-d, and a chip layout similar to that of *Fainman, et. al*o1 will be used, where the fluidic channels are bent at 90 degrees, allowing for efficient coupling of light into the liquid channel waveguide (also the microfluidic channel) (see Figure 5g). Light in the microfluidic channel produces an evanescent tail which overlaps well with the lipid monolayer, yielding high photocleaving efficiency per unit of optical power. Two additional layers of PDMS will be fused on top of the chip to allow for efficient pneumatic pump coupling.

Data Acquisition and Analysis

The injection performance of the device will be quantified. Oil consisting of n-octadecane will be streamed through the top channel and DI-water will be streamed through the bottom channel. The top channel will run at a pressure that creates a 1 cm/s flow to negate molecular diffusion traveling against the flow. The bottom channel will be run at a slightly larger pressure than the top channel to create a slight meniscus bulge in the direction of the aqueous phase. The fluid flow will be run for 10 minutes and then the aqueous phase will be collected at the out-port as a control sample. The control sample will be probed for the formation of emulsions using the DLS[32] and for stray lipids using NMR spectroscopy. These readings will be compared against pure DI water.

For the characterization of device performance, light will be pulsed with durations ranging from seconds to nanoseconds. Each sample will be given one pulse and a ten-second flow before collecting the liquid at the out-port. The fluid will be analyzed in the UV-vis to check the concentration of the nucleosides in solution[33]. NMR spectroscopy will be used to confirm that the solution contains pure nucleosides.

By knowing the concentration of nucleosides in solution and the volume of the collected sample, the total moles of nucleosides in the sample may be calculated. By dividing this value by the pulse time (with some extra considerations), the moles of dNTP released per second per unit intensity may also be calculated. From this data, a quantitative picture about the temporal release of molecules into the aqueous phase can be gathered.

Budget Estimate

Below is a safe budget estimate that allows for errors and extra research and development costs during the fabrication process. Note that many of the expensive costs may be offset by collaborating with other labs. The total cost includes the cost of the items that may be potentially borrowed from other labs.

Figure 6

Cost	Cleanroom	Materials and Tools	Chemical Reagents	Metrology	Unexpected Costs	
2100	Photolithography Process: 70 hrs					
500	E-beam Lithography: 5 hrs					
50		PDMS Kit				
6000 (or borrow)		Pneumatic Pressure Controller[34]				
5000 (or borrow)		Chemistry Process Equipment				
4000			Chemical Tab (Upper Bound)			
150				DLS: 7 hrs		
250				NMR: 7 hrs		
210				UV-Vis: 7 hrs		
1000				SEM: 20 hrs		
400				AFM: 20 hrs		
4000					Retries/Refabrications	
2000					Materials and Tools	
Total: $25,660						

Conclusion

Although the benefits of the photocleavable gate were discussed in the introduction, more comments must be made about the design of the device and the project as a whole.

As for the device, it is very scalable in terms of the number of independent reagents that may be gated within the same reaction zone. Multiple nanochannel arrays, each linking the main aqueous microfluidic channel to different oil channels, may be individually addressed by the same frequency of light without triggering the release of chemicals in other nearby arrays. As long as nanoplasmonic waveguides are used to confine the light in sub-wavelength regimes, individual nanochannel arrays may be dosed without crosstalk.

For the project as a whole, there are many moving parts. The mechanics of microfluidics, non-covalent surface forces, biomolecular membranes, cleanroom synthesis, and synthetic chemistry all come into play. In completing this project, it is expected that researchers familiar with these areas will have to be consulted.

No doubt it is a big undertaking, but these special gates may usher in a new class of devices capable of sending chemical signals as fast as their analytes may be probed. A new type of chemistry that operates on custom rate kinetics would be born.

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